Grade 6 Science – Mid-Term Worksheet

Science · Sara Elewa · Knowledge City Schools · Answer key ✅

Q1: Scientific terms

💡 Scientific terms: learn each definition with its term.

1. Write the scientific term: A huge ball of burning gases that releases energy.

A Desublimation
B Planet
C Star ✓
D Evaporation

Why: «Star»: A huge ball of burning gases that releases energy.

2. Write the scientific term: The system that consists of the sun and the planets.

A Astronomical unit
B Freezing
C Moon
D Solar system ✓

Why: «Solar system»: The system that consists of the sun and the planets.

3. Write the scientific term: A large round body of rocks or gases that moves around the sun.

A Desublimation
B Chemical bond
C Planet ✓
D Moon

Why: «Planet»: A large round body of rocks or gases that moves around the sun.

4. Write the scientific term: A unit used to measure the distances outside the solar system.

A Moon
B Planet
C Evaporation
D Light year ✓

Why: «Light year»: A unit used to measure the distances outside the solar system.

5. Write the scientific term: A unit used to measure the distance inside the solar system.

A Evaporation
B Compound
C Astronomical unit ✓
D Solar system

Why: «Astronomical unit»: A unit used to measure the distance inside the solar system.

6. Write the scientific term: A molecule with more than one type of atoms.

A Solar system
B Melting
C Condensation
D Compound ✓

Why: «Compound»: A molecule with more than one type of atoms.

7. Write the scientific term: A solid material with atoms that are arranged in a certain shape.

A Desublimation
B Evaporation
C Planet
D Crystal ✓

Why: «Crystal»: A solid material with atoms that are arranged in a certain shape.

8. **Write the scientific term: An attraction between atoms or molecules.**

A Chemical bond ✓ B Solar system
C Light year D Evaporation

Why: «Chemical bond»: An attraction between atoms or molecules.

9. **Write the scientific term: Changing of matter from the solid state to the liquid state.**

A Melting ✓ B Solar system
C Desublimation D Moon

Why: «Melting»: Changing of matter from the solid state to the liquid state.

10. **Write the scientific term: Changing of matter from the solid state to the gas state.**

A Astronomical unit B Planet
C Sublimation ✓ D Moon

Why: «Sublimation»: Changing of matter from the solid state to the gas state.

11. **Write the scientific term: Changing of matter from the liquid state to the solid state.**

A Freezing ✓ B Astronomical unit
C Moon D Crystal

Why: «Freezing»: Changing of matter from the liquid state to the solid state.

12. **Write the scientific term: Changing of matter from the liquid state to the gas state.**

A Desublimation **B Evaporation ✓**
C Chemical bond D Freezing

Why: «Evaporation»: Changing of matter from the liquid state to the gas state.

13. **Write the scientific term: Changing of matter from the gas state to the liquid state.**

A Condensation ✓ B Freezing
C Desublimation D Chemical bond

Why: «Condensation»: Changing of matter from the gas state to the liquid state.

14. **Write the scientific term: Changing of matter from the gas state to the solid state.**

A Light year B Evaporation
C Star **D Desublimation ✓**

Why: «Desublimation»: Changing of matter from the gas state to the solid state.

Q2: Choose the correct answer

💡 Review the lesson notes on the solar system, Earth's layers, resources, properties of matter and states of matter.

15. The solar system formed from a large cloud of gas and dust called a _____.

A galaxy
B nebula ✓
C comet
D crater

Why: A nebula is the cloud of gas and dust that formed the solar system.

16. According to the nebular hypothesis, why did the material begin to gather together?

A Gravity ✓
B Sound
C Electricity from Earth
D Ocean currents

Why: Gravity pulled the gas and dust together.

17. Which material was more likely to remain solid close to the young Sun?

A Rock and metal ✓
B Ice only
C Water vapor only
D Hydrogen only

Why: Close to the hot young Sun only rock and metal stayed solid; ice melted.

18. What is an asteroid?

A A large star
B A rocky object orbiting the Sun ✓
C A moon of Earth
D A cloud of gas

Why: An asteroid is a rocky object that orbits the Sun.

19. Which unit is most appropriate for measuring the distance between planets?

A Millimeter
B Astronomical unit ✓
C Gram
D Celsius

Why: Astronomical units measure distances inside the solar system.

20. What causes day and night on Earth?

A Earth's rotation ✓
B Earth's revolution
C The Moon's rotation
D Moon revolution

Why: Earth spins (rotates) on its axis → day and night.

21. Which Earth layer is the thinnest, solid outer layer of Earth?

A Outer core
B Mantle
C Crust ✓
D Inner core only

Why: The crust is the thin, solid outer layer of Earth.

22. Which of these is a natural resource?

A wood from a tree ✓
B nylon rope
C plastic bottle
D synthetic rubber

Why: Wood comes from nature; nylon, plastic and synthetic rubber are man-made.

23. Which material is usually synthetic?

A cotton
B wool
C polyester ✓
D stone

Why: Polyester is man-made (synthetic); cotton, wool and stone are natural.

24. Which is a renewable resource?

A coal
B sunlight ✓
C petroleum
D natural gas

Why: Sunlight is renewable; coal, petroleum and natural gas run out.

25. The amount of mass in a given volume is called:

A volume
B conductivity
C density ✓
D flexibility

Why: Density = mass ÷ volume.

26. Sugar dissolving in water demonstrates:

A solubility ✓
B magnetism
C hardness
D texture

Why: Sugar dissolving shows solubility.

27. Which material is a good electrical conductor?

A rubber
B dry wood
C copper ✓
D plastic

Why: Copper (a metal) conducts electricity well.

28. All matter is made of tiny particles called:

A cells only
B particles such as atoms and molecules ✓
C light rays
D sound waves

Why: Matter is made of atoms and molecules.

29. An atom is:

A a tiny building block of matter ✓
B a type of energy
C a liquid
D a mixture of substances

Why: An atom is the tiny building block of matter.

30. A molecule forms when:

A materials disappear
B two or more atoms bond together ✓
C a solid melts
D a liquid freezes

Why: A molecule forms when two or more atoms bond together.

31. In a pure substance, particles are:

A always different kinds mixed randomly
B the same type of particle ✓
C not moving at all
D only found in gases

Why: A pure substance has only one type of particle.

32. Which state has a definite shape and a definite volume?

A solid ✓
B liquid
C gas
D all fluids

Why: Solids have a definite shape and volume.

33. A liquid:

A has a fixed shape
B takes the shape of its container ✓
C cannot flow
D has no volume

Why: Liquids take the shape of their container but keep their volume.

34. Gas particles are generally:

A packed tightly in fixed positions
B close together and sliding past one another
C far apart and moving freely ✓
D not moving

Why: Gas particles are far apart and move freely.

35. The change from a solid to a liquid is called:

A freezing
B condensation
C melting ✓
D evaporation

Why: Solid → liquid is melting.

36. The change from a solid to a gas is called:

A condensation
B melting
C sublimation ✓
D freezing

Why: Solid → gas is sublimation.

Q3: Short answers

💡 Short-answer tip: remember the model answers from the worksheet.

37. Why can Earth support life better than most other planets?

A It contains oxygen in its atmosphere and freshwater ✓
B It is the biggest planet
C It is the closest planet to the Sun
D It has the most moons

Why: Earth has oxygen in its atmosphere and fresh water, so it can support life.

38. What is rotation?

A Spinning of an object around its axis ✓
B An object moving around another object
C Melting of ice
D The pull of gravity

Why: Rotation = spinning around its own axis.

39. What is revolution?

A Spinning of an object around its axis
B An object moving around another object ✓
C A change of state
D A type of rock

Why: Revolution = moving around another object.

40. Which pair lists two renewable resources?

A Sunlight and wind ✓
B Coal and oil
C Natural gas and coal
D Oil and natural gas

Why: Sunlight, wind and water are renewable.

41. Which pair lists two non-renewable resources?

A Sunlight and water
B Wind and sunlight
C Natural gas and coal ✓
D Water and wind

Why: Natural gas, oil and coal are non-renewable.

42. Which pair lists two natural materials?

A plastic and steel
B wood and copper ✓
C paper and plastic
D steel and paper

Why: Wood, copper and gold are natural materials.

43. Which pair lists two synthetic materials?

A wood and gold
B copper and wood
C plastic and steel ✓
D gold and copper

Why: Plastic, paper and steel are synthetic (man-made).

44. What is mass?

A The amount of matter in an object ✓
B The amount of space an object takes up
C The weight of water
D The speed of an object

Why: Mass is the amount of matter.

45. What is volume?

A The amount of matter in an object
B The amount of space taken by matter ✓
C The temperature of an object
D The hardness of an object

Why: Volume is the amount of space taken by matter.

46. A block has a mass of 60 g and a volume of 20 cm³. What is its density?

- **A** **3 g/cm³ ✓**
- **B** 40 g/cm³
- **C** 80 g/cm³
- **D** 1200 g/cm³

Why: D = M ÷ V = 60 ÷ 20 = 3 g/cm³.

47. Why are metals used for cooking pans, while plastic or wood are used for handles?

- **A** **Metals conduct heat; plastic and wood do not ✓**
- **B** Metals are lighter than wood
- **C** Plastic melts faster than metal
- **D** Wood conducts heat better than metal

Why: Metals conduct heat; plastic and wood do not, so handles stay cool.

Q4: Match the term

💡 Atom = basic unit of matter; Flammability = ability to burn; Property = a special quality; Pure substance = always the same composition.

48. A special quality or characteristic of something:

- **A** Atom
- **B** Flammability
- **C** **Property ✓**
- **D** Pure substance

49. Matter that always has the same composition:

- **A** Atom
- **B** Flammability
- **C** Property
- **D** **Pure substance ✓**

50. The basic unit of any matter:

- **A** **Atom ✓**
- **B** Flammability
- **C** Property
- **D** Pure substance

51. The ability of a type of matter to burn easily:

- **A** Atom
- **B** **Flammability ✓**
- **C** Property
- **D** Pure substance

Q5: Complete

💡 Polar (positive and negative sides), surface tension, carbon, chemical formula, element, compound, liquid, solid, gas.

52. Water is a _____ molecule because it has positive and negative sides.

- **A** Chemical formula
- **B** Gas
- **C** Surface tension
- **D** **Polar ✓**

53. An insect can walk on a pond because of the _____ of water.

A Gas
B Surface tension ✓
C Compound
D Liquid

54. A diamond and graphite are from the same element, that is _____.

A Element
B Gas
C Liquid
D Carbon ✓

55. H_2O is a _____ for water.

A Solid
B Liquid
C Gas
D Chemical formula ✓

56. Oxygen is an _____ found in compound insulin.

A Chemical formula
B Polar
C Element ✓
D Gas

57. Oxygen is an element found in _____ insulin.

A Carbon
B Solid
C Compound ✓
D Element

58. The substance that has fixed volume and takes the shape of a container is a _____.

A Liquid ✓
B Solid
C Carbon
D Element

59. The substance that has a fixed shape and volume is a _____.

A Polar
B Atom
C Solid ✓
D Surface tension

60. The substance that doesn't have a fixed shape or volume is a _____.

A Gas ✓
B Solid
C Polar
D Surface tension

Q6: Physical or chemical property?

💡 Physical properties can be seen or measured without changing the substance (colour, hardness, state). Chemical properties appear when the substance changes into something new (burning, reacting with air).

61. Sodium has a silvery-white color. This is a...

A Physical property ✓
B Chemical property

Why: Colour is a physical property.

62. Sodium is easy to cut with a knife. This is a...

A Physical property ✓ B Chemical property

Why: Being easy to cut (softness) is a physical property.

63. Sodium is solid at room temperature. This is a...

A Physical property ✓ B Chemical property

Why: Being solid at room temperature is a physical property (state).

64. Sodium burns with an orange flame. This is a...

A Physical property **B Chemical property ✓**

Why: Burning is a chemical change → chemical property.

65. Sodium becomes coated with a white substance when exposed to air. This is a...

A Physical property **B Chemical property ✓**

Why: Reacting with air to form a new white substance is a chemical property.

Q7: Chemical formulas

💡 Count every atom in the formula: H_2O = 2 H + 1 O = 3; NH_3 = 1 N + 3 H = 4; CO_2 = 1 C + 2 O = 3; CH_4 = 1 C + 4 H = 5.

66. Chemical formula of water:

A H_2O ✓ B CO_2

C NH_3 D CH_4

67. Number of atoms in a water molecule (H_2O):

A 2 **B 3 ✓**

C 4 D 5

68. Chemical formula of ammonia:

A H_2O **B NH_3 ✓**

C CO_2 D CH_4

69. Number of atoms in an ammonia molecule (NH_3):

A 3 **B 4 ✓**

C 5 D 2

70. Chemical formula of carbon dioxide:

A CO **B CO_2 ✓**

C C_2O D CH_4

71. Number of atoms in a carbon dioxide molecule (CO_2):

- A 2
- **B 3 ✓**
- C 4
- D 5

72. Chemical formula of methane:

- **A CH_4 ✓**
- B NH_3
- C CO_2
- D H_2O

73. Number of atoms in a methane molecule (CH_4):

- A 3
- B 4
- **C 5 ✓**
- D 6